

Q1.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 1, Analysis 3 Knowledge/Understanding: 1 mark for: Understanding or definition of a concentration ratio (1) Application: 1 mark for reason e.g. Change in shopping habits which has led to consumers shopping more frequently in discount stores/online shopping from other retailers (1) OR The smaller discount stores such as Aldi and Lidl have increased their market share (1) Analysis: Award further marks for linked developments e.g. because Aldi and Lidl focus on low prices so their sales have risen faster than those of the big four (1) more convenient to shop at local stores (1) The four firm concentration ratio has decreased by 3.6 percentage points (2) OR Calculation of the four firm concentration ratio in 2010 ($30.8 + 17.6 + 16.1 + 11.6 = 76.1\%$) (1) Calculation of the four firm concentration ratio in 2015 ($28.6 + 16.5 + 16.5 + 10.9 = 72.5\%$) (1)	(5)
Question Number	Indicative content	Mark
(b)	Knowledge 2, Application 2, Analysis 4 • Definition of monopsony • Food prices likely to fall Effects on food suppliers might include: • Financial problems from delays in receiving payments from supermarkets (Extract A) • Lower prices mean a reduction in revenue and profits • Increased costs due to paying excessive amounts for packaging and promotions. • Losses may even be made • Pressure on suppliers to cut costs • Fall in employment • Some food suppliers may exit the industry • Less funds for investment • Risk that suppliers will reduce product quality • Figure 2 indicates some supermarkets abused their market power e.g. Tesco and Morrisons. Effects on consumers might include: • Lower prices mean higher consumer surplus • Impact on food waste • Impact of cost cutting measures on quality of food and consumer choice. NB for a Level 3 response, candidates must consider both food suppliers and consumers in their answer. Answers must refer to the context.	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Answer	Mark
(b) continued	Evaluation 4 • Food suppliers may be able to increase efficiency and so remain profitable and stay in the industry. • Supermarkets may be reluctant to use their monopsony power for fear of GCA, government investigation and possible fines up to 1% of sales revenue. • Consumers may not benefit since supermarkets may not pass on lower costs in the form of lower prices. • Supermarket sector may not be a monopsony as there is more than one buyer –so perhaps less power over food suppliers. • According to Figure 2 some supermarkets may not abuse their position e.g. Aldi and Sainsbury. • Long term implications: exit of firms may risk supplies to consumers. • Consideration of bilateral monopoly e.g. large food supply firms or cooperatives. • Food suppliers may diversify e.g. farm shops. • Food suppliers may undertake vertical integration.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.



Question Number	Answer	Mark
(c)	Knowledge 2, Analysis 2, Application 2, Evaluation 2 Knowledge/understanding: 2 marks for identification of two measures (1+1) Analysis: 2 marks for linked explanation of these measures (1+1) Application: 2 marks for reference to data (1+1), e.g. • Fines - GCA can fine supermarkets up to 1% of annual revenue (1) • Regulation to ensure payments made within standard time (1) • Price regulation e.g. minimum pricing – to ensure fair prices are paid for supplies (1) /relevant diagram (1) • Promote new entrants and further competition in supermarket sector (1) • Block mergers if it leads to increased monopsony power (1) Evaluation: 2 marks for two evaluative comments, OR 2 marks for identification and linked development e.g. • Asymmetric information may lead to regulator making incorrect decisions (1) e.g. too lenient or too harsh on supermarkets (1) • Regulatory capture may arise (1) • Regulator may lack resources to investigate supermarket sector thoroughly (1) • Problems of gaining accurate information from suppliers (1) as they are reluctant to complain (1) • CMA may allow mergers (1) which increase monopsony power of supermarkets (1) • Government may counteract monopsony power by protecting farmers' income e.g. subsidies (1+1) • Problems of minimum price e.g. excess supply/food waste (1)	(8)

Question Number	Indicative content	Mark
(d)	Knowledge 2, Application 2, Analysis 2 • Identification of information gaps as a cause of market failure Information gaps as main cause of food waste e.g. • Food suppliers lack market information: e.g. predicting size of harvests – dependent on climate/ pest control and disease/ time lags in growing crops and raising livestock. • Supermarkets lack market information: e.g. predicting household demand – depends on tastes and fashion which may change/ population and migration changes/ impact of competitors. • Consumers lack market information: e.g. uncertainty over how to store food or whether food is fit for consumption after sell-by dates. Irrational behaviour as main cause of food waste e.g. • Computation issues e.g. consumers perceived demand may exceed actual demand for food / difficulty in comparing the different size of food products to fit with the actual amount of food required. • Habitual behaviour e.g. consumers may have a tendency to purchase more food than required/ better to have too much than too little food. • Influence of other people's behaviour e.g. consumers may purchase food that they might not want. • Impulse buying e.g. due to advertising or use of special offers. NB These points could count as Evaluation and the Evaluation points as KAA. NB A Level 3 response would typically consider both 'information gaps' and 'irrational behaviour'. NB There may be overlap between information gaps and irrational behaviour as possible causes of food waste.	(6)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(d) continued	Evaluation 4 - Consideration of relative proportions of food waste e.g. supermarkets appear favourable compared to food suppliers and households. - Other factors could cause food waste e.g. - Misleading advertising by firms so consumers purchase food they do not want. - Transport and storage problems due to road congestion / problems of importing food from overseas. - Consideration of whether it is rational to assume that consumers seek to maximise utility in allocating their expenditure / and firms seek to maximise profits in making production decisions. - Long run information gaps may narrow and irrational behaviour may reduce e.g. improved labelling and social enterprise, advice to customers on how to store food and use leftovers or use of new technology to extend freshness of food. NB These points could count as KAA and the KAA as evaluation.	(4)

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Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Answer	Mark
(e)	Knowledge 3, Application 3, Analysis 3 Problems of merger include: • CMA may block this horizontal merger since the combined firm has 27.4% market share/ exceeds the 25% legal monopoly figure. • Costs to the firms: possible diseconomies of scale such as overlap between location of stores/ co-ordination of staff/ redundancies/ implementation of new IT system (different types of diseconomies of scale may be awarded separately). • Impact of merger on consumer loyalty: the new store may not deliver consumer loyalty, for example, different brands between stores. • Lack of synergies and cultural clash • Underlying market conditions remain: consumers changing their shopping habits and merger does not address this issue. • Impact on food suppliers: increased monopsony pressure on suppliers may lead to reduced choice of food suppliers. An increased risk of GCA investigation. • Consideration of problems of other mergers and acquisitions: for example, BT takeover of EE. NB for a Level 3 response there must be reference to the context e.g. Figure 1 or Extract C.	(9)

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	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7–9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 - There may be benefits which offset the problems, such as - different types of economies of scale e.g. bulk purchase - rationalisation of stores and employees could lead to increased efficiency and profitability - increased market power - CMA could permit merger as sufficient competition still exists in the sector from online shopping and the growth of discount stores. - Prioritisation of problems with justification, for example, a lot of money could be spent on the proposed merger which is then blocked by CMA. - Short-run and long run implications: enormous upheaval in short run but sufficient cost savings could be made in long run to make it worthwhile.	(6)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially-developed chain of reasoning.
Level 3	5–6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Q2.

Question Number	Answer	Mark
(a)	Knowledge 2, Application 2, Analysis 1 Knowledge: 1 mark for identification of a possible reason and 1 mark for its relevance to a fall in the value of the currency. Possible reasons: • Fall in demand for peso (1) with reason (1) • Increase in supply of peso (1) with reason (1) • Current account balance: ◦ Fall in the value of X (1) because export prices have fallen (1) ◦ Rise in the value of M (1) because of economic growth (1) • Decrease in interest rates (1) in Chile relative to other countries (1) • Fall in FDI (1) with reason e.g. reduced foreign investment in Chilean mining (1) • Change in hot money flows (1) with reason e.g. speculative selling of peso (1) • Rise in value of other countries' exchange rates (1) with a change in relative rates (1) • Relatively high inflation in Chile (1) compared to other countries with which it trades (1)	
	Application: 2 marks for data references e.g. • Copper prices have fallen, Fig. 1 (1) e.g. to \$5000 (1) • The Chilean peso has decreased in value (1) • Calculation based on data (award a range of answers because the data is given for the whole year) e.g. 0.2 to $0.14 = 30\%$ fall (or similar) (1 + 1) • Inflation 5% in Chile (1)/above 2-4% target (1)/ e.g. because of higher unit labour costs (implied by high truck drivers' wages, Extract B) (1) • Fall in FDI as mining dries up, therefore reduced demand for peso • Application of an exchange rate diagram e.g. with peso on vertical axis with shift (1) Analysis: • 1 mark for explanation of the process linking to the cause to the identified change e.g. fall in copper prices means that the value of exports fall because demand is price inelastic (1) or imports rise in value because of economic growth sucking in imports (1) diagram showing shifts in demand or supply of pesos linked to correct explanation (1) NB only award 1 mark in total for use of a diagram	(5)

Question Number	Answer	Mark
(b)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 NB an answer might provide positive externalities as KAA and negative externalities as evaluation NB do not award effects on copper mining firms, unless an indirect effect. NB an answer which does not relate both to firms and communities is limited to 5/6 KAA marks. NB do not award externalities themselves, e.g. pollution, the environment. Award the <i>impact</i>. Knowledge and Analysis Knowledge/ implicit understanding of externalities (1) e.g. knowledge that externalities are an effect on third parties, or outside (or not accounted for by) the market mechanism, or the difference between private and social costs or benefits Firms (in businesses other than mining) Points showing the impact on firms might include: • farms lacking water because mining companies are using it (1+1) • high demand for energy means that other firms will face high energy costs (1+1) • high exchange rate can cause problems for other exporting firms (1+1)	

	Communities Points showing the impact on communities might include: • problems of water access for households (1+1), • large-scale transport causes noise, air pollution (1+1), • people have to move away to find work (social dislocation) (1+1) • impact on employment and incomes as mining declines (1+1) • impact on firms that supply the mining sector (1+1) Some impacts could be seen as firms and communities, e.g. impact on wider economy of higher wages as firms have to compete with unionised highly paid copper miners for employment. Application: • Externality diagram illustrating negative or positive impact using marginal analysis (up to 2 marks: 1 mark for correct MSC and MPC drawn, 1 mark for correct equilibrium shown/deadweight loss) • Water has to be pumped 200km (Extract B paragraph 4) which uses fossil fuels/is noisy/ugly/affects tourist areas • Mines are deeper and use more fuel (carbon emissions have impact on climate and therefore on welfare of many) • Desalination plants might affect other firms or consumers (this might be negative or positive)	
	Evaluation (2 marks for any relevant point, two points 1 +1): • Positive externalities, e.g. ◦ impact of local multiplier on economy, improved infrastructure affects other industries ◦ Improved wages and working conditions for workers (in businesses other than mining) ◦ The copper mining has already helped to fund improvements in welfare, such as the financial system and pensions ◦ Benefits of higher value of currency, e.g. terms of trade improvement • Difficult to know how large the externalities are or who is bearing the cost • The benefits might outweigh the costs, e.g. reduction in poverty, improved fiscal powers, strength of financial markets, improved local infrastructure • Other standard evaluation points, e.g. Extract B a drought has made the problem more severe, time to acquire environmental impact assessments (Extract B paragraph 5) might not be a reliable indicator	(8)

Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 4 • Link between falling copper prices and falling incomes and government revenue e.g. dramatic loss of government revenue from peak of \$11.5 billion • Exposure to risk e.g. high proportion of GDP and exports dependent on copper so this could cause risk of falling incomes • Price volatility e.g. might cause reduced FDI/investment • Impact of declining terms of trade might be used • Successful exports of copper might force up the exchange rate, other 'Dutch disease' issues e.g. impact on non-copper sector • Falling demand for copper might cause peso to fall in value, causing declining terms of trade/cost push inflation • May use diagram to illustrate effects of low PES, PED • Award use of concepts or theories such as Prebisch-Singer, with link to terms of trade and changing world incomes. • Firms may close down e.g. Extract A lines 8-9: award a microeconomics diagram showing shutdown point NB Do not award any parts of answers which are related to externalities A KAA point referring to changes in tax revenues can only be awarded Level 3 if a specific link is made to the source of the tax revenue e.g. – corporation tax or profits of a state-owned enterprise For a Level 3 answer there must be reference to dependency rather than just falls in copper prices, and the wider economy rather than just copper mining firms	(8)

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	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 - Advantages of copper dependency for Chile's economy: - Copper dependency is an advantage when copper prices are high e.g. they have been high for a prolonged period, with many advantages - Use of data to support this, e.g. \$11.5 bn a year tax revenue, or percentage changes in prices Figure 1 - Low national debt, falling levels of poverty, inward direct investment - Other industries thriving e.g. salmon industry - Chile's economy 'best run in the region' and has financial strength to overcome copper price changes, relative to other countries who are dependent on primary products e.g. Venezuela - The fall in the exchange rate corrects the problem of over dependency - Copper prices are far less volatile than 2007–2011	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to the context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 16 marks for KAA, for effects of policies: Microeconomic effects may include: • fiscal, e.g. 'mildly expansionary policies' such as cutting income tax will increase real wages • monetary e.g. cutting interest rates can reduce the costs of borrowing for individuals and firms. • supply side policies, e.g. privatisation, free trade can make markets more competitive • growth and development strategies e.g. education, debt relief etc. can further alleviate poverty • price controls Macroeconomic effects may include: • Cutting interest rate might cause the peso to further decrease in value, which would cause inflation / increase the value of exports leading to current account improving • Privatisation or other supply side policies can increase international competitiveness NB for a Level 4 response there must be reference to both economic growth (e.g. rising incomes) and economic development (e.g. longer life expectancy through improved living standards) and a focus on effects for Level 3 and above. 9 marks for evaluation – might include: • Expansionary fiscal policies might involve an improvement in the redistribution of income • Objectives might conflict, e.g. cutting interest rates can cause inflation or widen inequality, or increased government spending to produce growth might cause inflation (e.g. Phillips Curve) • Policies might conflict, e.g. increasing length of time to gain a permit to reduce externalities can cause a negative supply side impact e.g. by raising production costs for firms.	(25)

Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays ability to apply knowledge in context but will focus on small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models.
Level 2	5–8	Shows ability to apply economic ideas and relate them to economic problems in context. Displays knowledge and understanding of economic principles, concepts and theories to make limited analysis or narrow analysis.
Level 3	9–12	Analysis is clear and coherent with evidence well integrated, although may focus on some of the broad elements of the question more than others. Shows ability to apply economic ideas and relate them directly to the broad elements in the question.
Level 4	13–16	Analysis is relevant, clear and coherent with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic effects. A clear understanding of economic principles, concepts, theories and arguments.

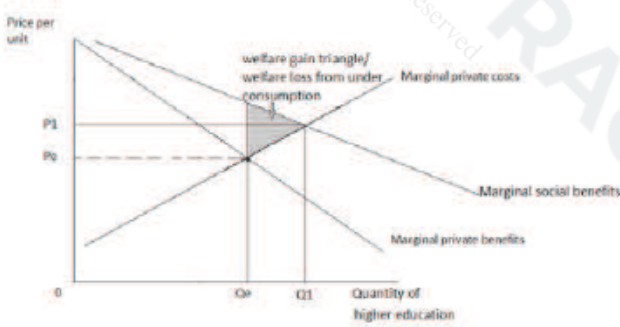
Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of evaluative comments without explanation.
Level 2	4–6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to the context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 This mark scheme is based on an increase in I, but answer can be given from the point of view of a decrease Up to 16 marks for effects: Microeconomic effects may include: • Effects on workers – higher incomes, more jobs • Effect on firms, e.g. increase in productivity, or for firms that produce capital goods, a change in demand • Increase in demand as consumer and business confidence improves • Improved quality of life as a result of research into and development of products Macroeconomic effects may include: • Shift (increase) in AD, with multiplier effects, increasing real GDP • Shift (increase) in AS increasing capacity e.g. through investment in education (Extract A paragraph 5) (LRAS shift only, may be implicit) • Inflation, if linked to expansionary macroeconomics policies • Balance of payments – e.g. effects of inward FDI: short term improvement on financial account, long term impact on current account Award Keynesian and Classical approaches with equal weight. The opposing viewpoints may be used as evaluation. NB a Level 4 answer refer to <i>changes in</i> investment rather than levels. This may be implicit e.g. through a fall in AD NB do not award for impact other than on Chile. Do not award for causes rather than effects.	
	Up to 9 marks for evaluation points include: • Investment in capital might decrease employment • Investment in human capital e.g. education in Extract A, is hard to measure in terms of effectiveness • Data points to both increases and decreases in investment. Other forms of injection into the circular flow might be more sustainable, e.g. G • Depends if the investment is internal or FDI as to the effects on the balance of payments • Depends on whether there is a demand deficiency (Keynesian viewpoint) or failings in the supply side	(25)

Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays ability to apply knowledge in context but will focus on small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models.
Level 2	5–8	Shows ability to apply economic ideas and relate them to economic problems in context. Displays knowledge and understanding of economic principles, concepts and theories to make limited analysis or narrow analysis.
Level 3	9–12	Analysis is clear and coherent with evidence well integrated, although may focus on some of the broad elements of the question more than others. Shows ability to apply economic ideas and relate them directly to the broad elements in the question.
Level 4	13–16	Analysis is relevant, clear and coherent with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic effects. A clear understanding of economic principles, concepts, theories and arguments.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of evaluative comments without explanation.
Level 2	4–6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7–9	Evaluative comments supported by relevant reasoning. Evaluation recognises different viewpoints and can be critical of the evidence provided.

Q3.

Question Number	Indicative content	Mark
	Knowledge 4, Application 4, Analysis 8, - Define imperfect information/ asymmetric information - Identification of asymmetric information in market for education - Not know contents of course- choose wrong course- affect ability to gain career/ enjoyment on course - Students not aware of the returns from education e.g. 2 A levels/ degree etc- may under consume/ over consume- missing out on external benefits - Not aware of external benefits of education to third parties- under consume leading to under consumption as shown in external benefit diagram - True cost of education- books, accommodation, tuition fees etc – may lead some to accrue debts they cannot afford - Information not known about some subjects- under consumption may be a problem- e.g. lack of languages teachers, engineers etc. 	(16)

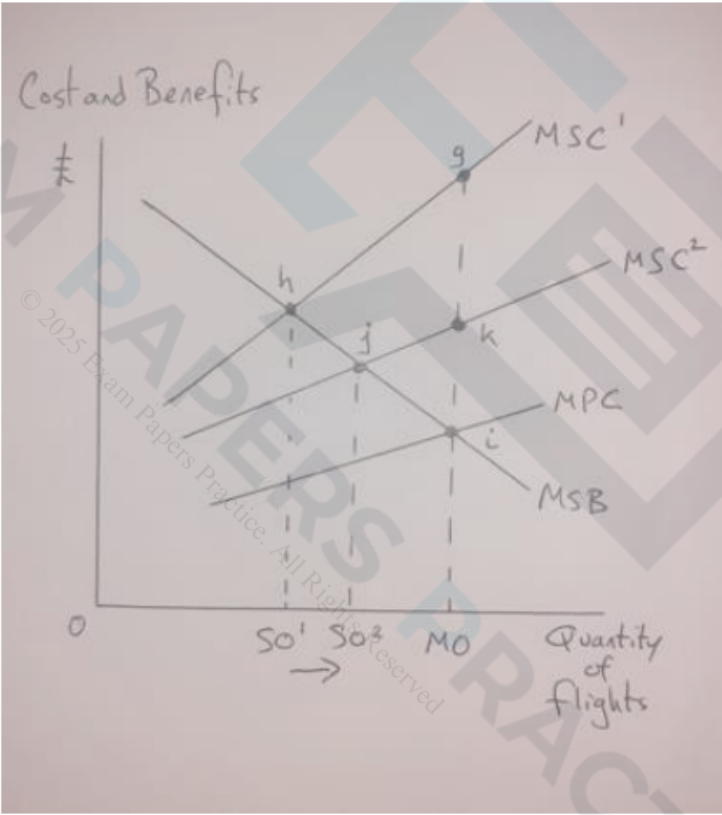
Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	5–8	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	9–12	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	13–16	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
continued	Evaluation 9 Evaluation - information available on internet/ by universities to reduce the problem - Long term- less of an issue as more go to university and friends and family can share information - Can we afford for all to access HE- budget deficit issues - Measurement of costs and benefits hard to quantify to know positions of curves	(9)

Q4.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	4–6	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Question Number	Answer	Mark
(a)	Knowledge 1, Application 2, Analysis 2 Knowledge and Analysis: up to 3 marks for e.g. - Staffing levels fall (1) - Diminishing marginal productivity means for an additional unit of labour the additional increase in productivity declines (1) so the additional cost of hiring more labour (1) needs to be set against the additional productivity gained (1) - As marginal productivity declines/diminishing returns set in (1) marginal cost increases at a faster rate (1) given labour is homogenous (1) - In the short run labour is variable (1) all other factors of production are held constant (1) so an additional unit of labour will result in less additional output (1) - An additional unit of labour results in: marginal product declines (1) average product declines (1) total product increases at a slower rate (1) total cost rises at a faster rate (1) marginal cost increases (1) average cost rises (1) Application: up to 2 marks for e.g. - lower marginal productivity of an additional crew member (1) explains why American Airlines are reducing staffing levels (1) - The first cabin crew member serves 50 passengers, the second serves 60 additional passengers, the third serves 30 more, the fourth serves 20 (1+1) - The number of additional meals served will decline per additional crew member (1) as crew members start to get in each other's way (1) - The available space on the plane is fixed (1) and seating/trolley/meals/doors available (1) - Lower cost per passenger (1) cheaper tickets (1)	(5)

Question Number	Answer	Mark
(b)	Knowledge 2, Analysis 2, Application 2, Evaluation 2 Knowledge/understanding: 2 marks (1+1) e.g. • Social optimum is when $MSC = MSB$ (1) • Social optimum position moves right (1) • Market equilibrium output (e.g. MO) position identified (1) • Downward sloping MSB (1) • $MSC = MPC + MEC$ (1) 	

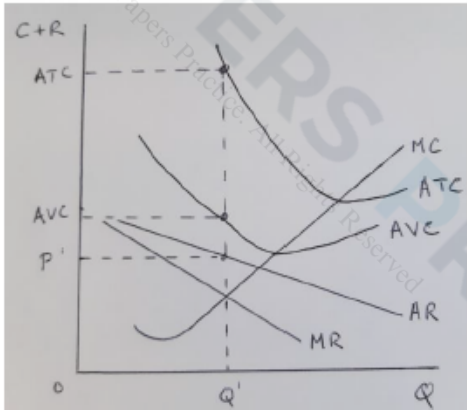
	Analysis: 1 mark for linked explanation of each identified benefit (1+1) e.g. • Third party effects are lower (1) so external costs are lower/MEC falls (gi to ki) (1) • MSC shifts/skews right (1) showing new social optimum position (MSB=MSC) (1) • Welfare loss falls (1) from ghi to kji (1) Application: 2 marks for reference to Extract B e.g. • Thomas Cook airline emissions are lower/switch fuel types/new engines (1) • In 2018 Thomas Cook Group Airline was included in the top 10 of the world's most fuel-efficient airline (1) • Lower insurance bills/clean up bills for companies/households at risk of flood damage/storms (1) loss of cattle/crops due to drought declines (1) • CO₂ emissions per passenger declines (1) quantity of flights increases (1)	
	NB These application points may be annotated on the diagram. Evaluation: 2 marks for two evaluative comments, OR 2 marks for identification and linked development of one evaluative comment e.g. • Difficulty in measuring size of negative externality (1) third party effects (1) • CO₂ emissions from travel agents, hotels and holiday activities increase/remain high (1+1) • Computational problems in measuring CO₂ emissions from planes (1) faulty industry data (1)	(8)

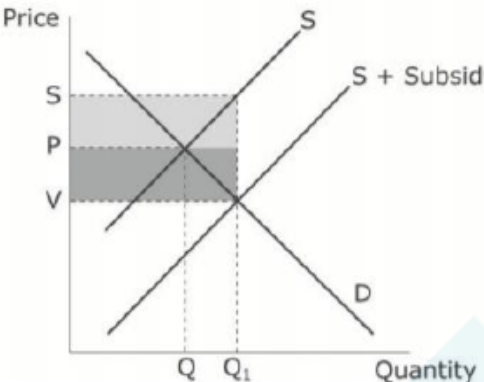
Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 2 Case for principal agent problem being significant. Conflict of interest between the principal (shareholder) and the agent (CEOs/directors/managers) creates problems for Thomas Cook: • Moral hazard – no consequences for the failings of management results in high-risk behaviour • Chief Executive incentivised by £500 000 bonus and £8.5 million salary possibly linked to sales or market share rather than the long-term profitability of Thomas Cook • To increase sales or market share the Chief Executive may have focussed on mergers with other Travel businesses, price wars with competitors or investing in buying new hotels and planes rather than keeping costs low and paying off debts • Chief Executive may have prioritised a greener image by reducing emissions or better pay to avoid pilots striking and sought to satisfy shareholders instead reducing the long-term profitability of Thomas Cook	(6)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no link between causes and consequences.
Level 2	3–4	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or the answer may lack balance.
Level 3	5–6	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are applied appropriately to the broad elements of the question.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 Case against 'principal agent problem' - Employee share-ownership schemes address the problem. £4m of the CEO's £8.5m earning is in shares – so as an employee he is motivated by profit maximisation, bringing him in-line with shareholders. His shares are now worthless - CEO worked "exhaustively" to rescue Thomas Cook and create a long-term strategy. By owning planes and hotels they would have assets and not borrow to rent. In the long-run Thomas Cook could have moved its sales more on-line - Shareholders have lost capital value but the CEO has lost his job, earnings and his reputation - Shareholders can hold the CEO accountable at their AGM, reject pay and bonus awards and place the CEO on a short-term contract to avoid the long-term contract short-term gain problem - Other factors that caused closure – fuel prices, competition.	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(d)	Knowledge 2, Application 2, Analysis 4 Case for subsidy: • Subsidy is financial aid, grant • In long run required more to cover AC • Subsidy may have lowered AC, may be shown as shift on diagram • Benefit of subsidy to producer may cover their AVC in short-run or AC in long-run. • Avoid government rescuing 600 000 travellers • Avoid government paying benefits to up to 21 000 employees and those losing jobs in related industries e.g. hotels • Subsidy may have leveraged help from banks and shareholders • Thomas Cook may have recovered - Package holidays may have come back into fashion – aging population. Recession may have seen a rise in demand for package holidays as an inferior good, YED negative • Diagram may be used, for example: 	

	OR Incidence of a Subsidy  NB a diagram is not required NB case against subsidy may be seen as KAA and case for as evaluation or vice versa.	(8)
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Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1-2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3-5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6-8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(d) continued	Evaluation 4 Case against subsidy Lack of finance £150 million not enough - £1.5 billion loss. - Needed help from bank loan (zombie-firm), shareholders or another company taking over Thomas Cook. Not willing to help. £1.5 billion loss in May 2017, already in long-run - Cheaper and more effective to nationalise to rescue employees and suppliers Merger with MyTravel - Diseconomies of scale, $LRAC > AR$ - Lack of synergy - Loss making business	
	Falling demand for Package holidays - AR shifting in - Booking online through competitors rather than Thomas Cook high street outlets - Package holiday inferior good, YED negative. - Climate change – stay at home. Opportunity cost Dependency Government lacks funds- budget deficit Moral hazard No consequences for the failings of management will teach other businesses who get into commercial difficulties that government will pick up the pieces NB: Case against subsidy may be seen as KAA and case for as evaluation or vice versa.	(4)
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(e)	Knowledge 3, Application 3, Analysis 3 Exploiting market power • Jet2 increases 'price per person' by £36 after Thomas Cook closes (Fig 2) • Thomas Cook, with a 13% market share in 2019, exits the market causing Jet2 to gain market share, possibly close to 25% - gaining monopoly power/duopoly/oligopoly (Fig 1) • Jet2 may be using its monopoly dominance to set a price above the market clearing/allocatively market price • Consumer surplus reduced, producer surplus gained. • With less competition for package holidays, Jet2's PED becomes relatively more price-inelastic resulting in greater price setting powers for package holidays • Jet2 may now be able to set its price at a profit maximising level of £824 rather than possible sales maximisation/revenue pricing/limit pricing/predatory pricing at £788 • Demand for Jet2 holidays increasing as a close substitute to Thomas Cook • Remaining companies colluding, use of game theory. N.B KAA may be used as EV and vice-versa.	(9)

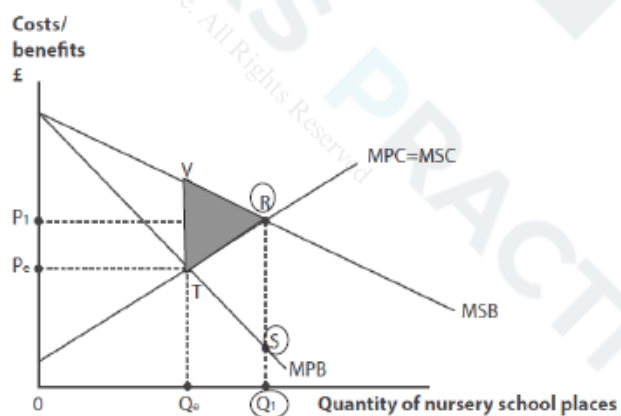
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–3	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	4–6	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	7–9	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(e) continued	Evaluation 6 Responding to market forces • Demand for Jet2 holidays increasing as a close substitute to Thomas Cook. This may have been happening before Thomas Cook closure • Thomas Cook exiting the package holiday may have removed a surplus of package holidays • At the price of £788 there is now a shortage of holidays to Sorrento, as shown by excess demand due to cancelled Thomas Cook holidays • Market forces resulting in Jet2 extending their supply at higher prices • As they increase the price to £824 the market may clear	(6)

	• Consumers may decide not to buy at higher price - to stay home or become anxious about firms closing and cancelled holidays • Other factors may have caused demand to rise or supply to shift inwards • Market share may not be reallocated to Jet2 • Jet2 is a distant substitute and may not go to same destinations as Thomas Cook • Companies do not collude as fear a whistle-blower and fines 10% of revenue • Rising marginal costs for Jet2 to reduce CO₂ emissions/ buy more fuel to new destinations/ costly take overs. NB KAA may be used as EV and vice-versa.	
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Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1-2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3-4	Evidence of evaluation of alternative approaches which is unbalanced. Evaluative comments with supporting evidence/reference to context and a partially-developed chain of reasoning.
Level 3	5-6	Evaluative comments supported by relevant chain of reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Q5.

Question Number	Answer	Mark
	Knowledge 1 Knowledge/understanding 1 mark for definition of external benefits e.g. • Positive externality (1) • When the social benefits of an economic action are greater than the private benefits (1) • External benefits = social benefits – private benefits (1) • Benefits accruing to a 3rd party (1) • Spillover effects (1) • Outside the price mechanism (1)	(1)
Question Number	Answer	Mark
	B	(1)
Question Number	Answer	Mark
	Application 2 Identify welfare gain area VTR (1) Annotate or identify the social optimum output as Q₁ or R or S (1) 	(2)

Q6.

Question Number		Mark
(a)	Knowledge 2, Application 2, Analysis 1 Knowledge: • 'Good' deflation - Caused by an increase in aggregate supply (1) • 'Bad' deflation - caused by falling aggregate demand (1) Application: 2 data references from Extract 1: (2 marks) • For 'good' deflation - falling price of unit of computing power (1) • For 'bad' deflation - recession in eurozone (1) Analysis: 1 mark for explanation of the link between: Either • AS and 'good' deflation: because it is caused by new technology; falling raw material prices which result in higher real output (1) OR • AD and 'bad' deflation: because it is caused by falling real wages which result in lower real output. (1)	(5)

Question Number		Mark
(b)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Knowledge and analysis: 1 mark for each reason, up to 2 marks (K). A further 2 marks for supply and demand diagram(s) or further development or explanation (A) - Fall in demand - recession in eurozone; slowdown in China and Asia - Increase in supply - shale oil Application (1+1 marks): Specific data reference from Figure 1, e.g. fall in oil price from around \$115 a barrel in mid 2014 (1) to just over \$60 a barrel by the end of 2014 (1) Specific data references from Extract A: e.g. 50% increase in oil supply in USA (1) Evaluation: (2 marks for any relevant point): - Prioritisation - discussion of which factor has had the greatest impact - Fall in demand might be more temporary than increase in supply.	(8)

Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 4 - Meaning of a cartel: group producers acting together; restricting supply to force up price - The producers act as a monopolist when a cartel is formed - All major producers must be members of the cartel for it to be able to succeed - No new substitutes for the product must be available (in this case shale oil is an effective substitute for oil) - Supply must be controlled – without ‘cheating’ by individual producers.	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1 - 2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3 - 5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6 - 8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 • Much will depend on relative price of oil in relation to sources of renewable energy • A cartel might be more successful in the short run than in the long run when alternatives may be developed and become more available • Less powerful members of the cartel are likely to cheat • Difficulty of controlling activities of major producers • Impact of increase in shale oil/gas production • Inability to prevent entry of new producers (discussion of the degree of contestability in the market.	(4)

Level	Marks	Descriptor
0	0	No evaluative comments.
1	1-2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
2	3-4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 Microeconomic effects: • on consumers - increase in real value of savings; falling prices so increase in consumers' surplus • on producers - lower input prices leading to falling costs of production and possibly higher profits • on workers - could result in nominal wage cuts • on shareholders - falling profits might result in falling dividends. Macroeconomic effects: • impact on Investment - less incentive for firms to buy capital equipment • impact on Employment - could lead to a 1930s type depression with massive unemployment • impact on Economic growth - could result in prolonged depression • impact on Real value of a country's national debt increases. Possible evaluation points include: • impact depends on the causes of deflation – whether it is 'good' or 'bad' • consumers face increase in real value of mortgages and debts • producers face falling revenues and profits • impact on macroeconomic variables (growth, employment, balance of payments) dependent on type of deflation.	(25)

Knowledge, application and analysis		
Level	Marks	Descriptor
	0	A completely inaccurate response.
Level 1	1 - 4	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	5 - 8	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	9 - 12	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	13 - 16	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Evaluation		
Level	Marks	Descriptor
	0	No evaluative comments.
Level 1	1 - 3	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	4 - 6	Evidence of evaluation of alternative approaches which is unbalanced, leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	7 - 9	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 Microeconomic effects: • impact on costs of production resulting in higher profits for businesses • impact on revenues and profits of oil companies (reference to price inelasticity of demand) • impact on investment decisions by firms in the oil market, especially with reference to shale oil • impact on wages of workers in oil industry • impact on consumer real incomes and on demand for other goods and services • impact on green energy industries (reference to cross elasticity of demand). Macroeconomic effects: • lower global rates of inflation (AD/AS analysis) • impact on global growth rates • impact on balance of payments on current accounts of oil importing countries and oil exporting countries • impact on CO₂ emissions: consideration of impact on climate change on different countries - reference to external costs. Possible evaluation points include: • impact on different industries will depend on their oil costs as a proportion of total costs • impact on households will depend on cost of oil in household budgets • impact on demand for other goods and services will depend on other factors e.g. state of world economy, consumer confidence, household debt • impact on inflation rates and growth rates depends on elasticities of AD and AS curves • impact on balance of payments on current account of different countries depending on whether they are net importers or net exporters of oil • impact on external costs depends on how different price elasticity of demand for oil.	(25)

Knowledge, application and analysis		
Level	Marks	Descriptor
	0	A completely inaccurate response.
Level 1	1 - 4	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	5 - 8	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	9 - 12	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relate them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	13 - 16	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Evaluation		
Level	Marks	Descriptor
	0	No evaluative comments.
Level 1	1 - 3	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	4 - 6	Evidence of evaluation of alternative approaches which is unbalanced, leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially developed chain of reasoning.
Level 3	7 - 9	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Q7.

Question Number	Indicative content	Mark
	Knowledge 4, Application 4, Analysis 8, Evaluation 9 • Definition of external costs. • Identification of external costs likely in the energy market. • Explain the reason for government intervention, e.g. overconsumption and the need to reduce it. • Diagram identifying external costs and the need to reduce it – linking to overproduction/deadweight loss: – Indirect taxation – ad valorem and specific – costs rise for firm – reducing supply and quantity/may use funds to compensate third party/increases incentive to move to production of energy with lower external costs. But if demand inelastic just passed on to consumer/little impact on consumer/measurement problem to decide size of tax/may not be reinvested/used to compensate third parties/avoidance/evasion – Tradable pollution permits – how it works/incentive to reduce pollution to be able to sell them/those that do rewarded/ those that don't have added costs/ But only works if right quantity sold/some may not bother if surplus permits/costs to administer/monitor – Extension of property rights – third party can seek compensation. But only those that can afford to pursue will/no guarantee that you can prove they are the guilty party/energy companies large have expensive and powerful lawyers – Regulation – banning or setting limits on energy production. But expensive to police/measurement problem – what limits to set. • Diagrams may be rewarded where appropriate. NB For a Level 4 response, candidates must consider two methods in their answer.	
		(25)

Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	5–8	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response or superficial, two stage chains of reasoning only.
Level 3	9–12	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to apply economic concepts and relates them directly to the broad elements of the question with evidence integrated into the answer. Analysis is clear and coherent, although it may lack balance. Chains of reasoning are developed but the answer may lack balance.
Level 4	13–16	Demonstrates precise knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using appropriate examples. Analysis is relevant and focused with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	4–6	Evidence of evaluation of alternative approaches which is unbalanced leading to unsubstantiated judgements. Evaluative comments with supporting evidence/reference to context and a partially-developed chain of reasoning.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Q8.

Question Number		Mark
(a)	Knowledge 2, Application 2, Analysis 1 Knowledge 1 mark for identification of a possible cause and 1 mark for its relevance to energy use Possible causes: • Increase in world population (1) • Increase in devices requiring electricity (1) • Increase in world GDP and living standards (1) Application: 2 marks for data references from Figure 1 (identifying the forecast change): • Demand for nuclear and renewable more than double • Demand for gas doubles • Demand for coal and oil increase by between 30% and 50% • Comment on increase in overall demand Analysis: • 1 mark for explanation of the link between the identified cause and energy use	(5)
(b)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Knowledge and analysis • Increasing injections (exports) (K) may lead to a boost to the circular flow and an increase in AD if not offset by greater leakages (A) • Decreasing leakages (imports) (K) may lead to a reduction in the circular flow and reduce AD (A) Application: • Exports increasing, imports decreasing, but until 2020, leakages still greater than injections so overall leakage from circular flow (2) • Or 2 specific data references from figure 4 Evaluation (2 marks for any relevant point): • These are forecasts so the impact on the circular flow may be different from that described above • Other injections and leakages influence the circular flow which may offset the changes described above	(8)

Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 4 - recognition that gas and solar energy/nuclear energy are substitutes - application of cross elasticity of demand: definition or formula value will be positive - use of information provided to draw conclusion (reference to first paragraph, Extract 2): a fall in the price of gas will lead to a fall in the demand for solar energy and nuclear energy.	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.
Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.

Question Number	Indicative content	Mark
(c) continued	Evaluation 4 • much will depend on relative price of gas in relation to solar and nuclear energy • recognition that there will be a time lag before any impact on market for other forms of energy because supply very price inelastic • investors will want to be certain that change in relative prices is long term • impact may be limited due to the rising demand for global energy	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/ reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to context. Evaluation recognises different viewpoints and/or is critical of the evidence.

Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 Microeconomic effects: • on price of gas to consumers – impact on consumers’ surplus • on price of gas to producers – impact on producers’ surplus • implications for producers of other forms of energy • external costs – on environment and water supply Macroeconomic effects: • on investment – multiplier effect on GDP so promoting economic growth • on employment – an estimate that 35 000 jobs will be created • balance of payments on current account e.g. competitiveness may increase if fracking results in lower energy prices leading to an improvement in exports. Also, less need to import energy from abroad • rate of inflation may fall – lower energy prices Possible evaluation points include: • Impact on price of energy may be less significant than in the USA • Impact on other forms of energy – much will depend on how much gas is obtained from fracking • Difficult to forecast the magnitude of external costs or to place a monetary value on them • Impact on macroeconomic variables (growth, employment, balance of payments) dependent on amount of gas found, the price at which it sells, changes in prices of other forms of energy	(25)

Knowledge, application and analysis		
Level	Marks	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays ability to apply knowledge in context but will focus on a small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models
Level 2	5–8	Shows ability to apply economic ideas and relate them to economic problems in context. Displays knowledge and understanding of economic principles, concepts and theories to make limited analysis or narrow analysis
Level 3	9–12	Analysis is clear and coherent with evidence well integrated, although may focus on some of the broad elements of the question more than others. Shows ability to apply economic ideas and relate them directly to the broad elements in the question.
Level 4	13–16	Analysis is relevant, clear and coherent with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic effects. A clear understanding of economic principles, concepts, theories and arguments.

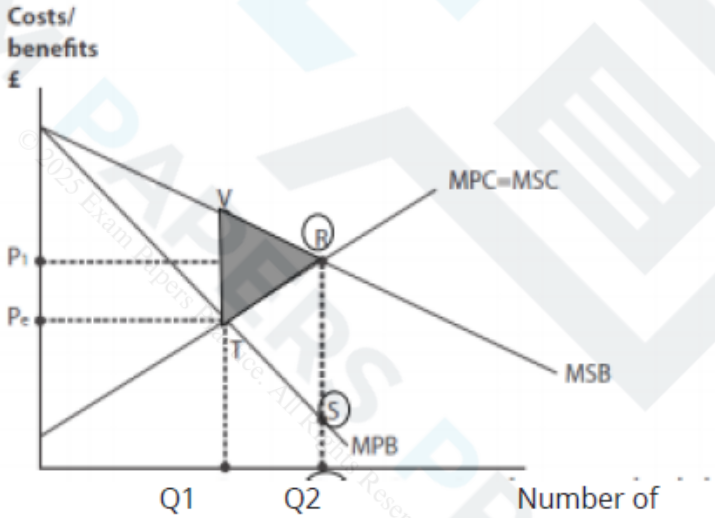
Evaluation		
Level	Marks	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of evaluative comments without explanation.
Level 2	4–6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7–9	Evaluative comments supported by relevant reasoning. Evaluation recognises different viewpoints and can be critical of the evidence provided.

Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8 Microeconomic effects: • impact on revenues and profits of energy companies • impact on investment decisions by firms in the energy market • impact on wages of workers in energy industries • impact on consumer real incomes and on demand for other goods and services. Macroeconomic effects: • lower costs of production • lower global rates of inflation (AD/AS analysis) • impact on global growth rates • impact on balance of payments on current account of different countries • increase in CO₂ emissions: consideration of impact on climate change on different countries, reference to external costs • impact on consumers and producers of the main countries supplying energy. Possible evaluation points include: • impact on different industries will depend on their energy costs as a proportion of total costs • impact on households will depend on cost of energy in household budgets • impact on demand for other goods and services will depend on other factors e.g. state of world economy, consumer confidence, household debt • impact on inflation rates and growth rates depends on elasticities of AD and AS curves • impact on balance of payments on current account of different countries depending on whether they are net importers or net exporters of energy • impact on external costs depends on how different energy sources are affected	(25)

Knowledge, application and analysis		
Level	Marks	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays ability to apply knowledge in context but will focus on a small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models
Level 2	5–8	Shows ability to apply economic ideas and relate them to economic problems in context. Displays knowledge and understanding of economic principles, concepts and theories to make limited analysis or narrow analysis
Level 3	9–12	Analysis is clear and coherent with evidence well integrated, although may focus on some of the broad elements of the question more than others. Shows ability to apply economic ideas and relate them directly to the broad elements in the question.
Level 4	13–16	Analysis is relevant, clear and coherent with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic effects. A clear understanding of economic principles, concepts, theories and arguments.

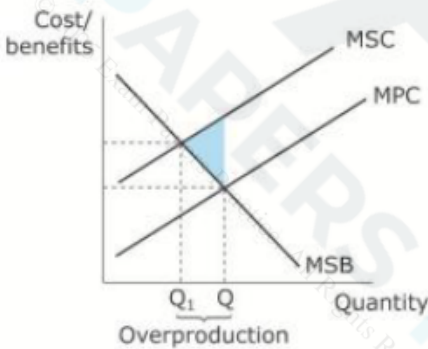
Evaluation		
Level	Marks	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of evaluative comments without explanation.
Level 2	4–6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7–9	Evaluative comments supported by relevant reasoning. Evaluation recognises different viewpoints and can be critical of the evidence provided.

Q9.

Question Number	Answer	Mark
(a)	Knowledge 2, Application 2 1 mark for the original market equilibrium where $MPB = MSC (=MPC)$ 1 mark for correct shift from MPB to MSB 1 mark for showing the external benefit V to T 1 mark for welfare gain identified VRT 1 mark for the new equilibrium identified where MSB equals MSC (R) or (S) or $Q2$ Social Optimum. Diagram required e.g.  ME SO	(4)

Question Number	Answer	Mark
(b)	Analysis 1 The only correct answer is C <i>A is not correct because an external benefit to a third party is not identified</i> <i>B is not correct because an external cost to a third party is not identified</i> <i>D is not correct because a private cost plus external cost to a third party is not identified</i>	(1)

Q10.

Question Number	Answer	Mark
(a)	Knowledge 2, Application 2 1 mark for the correct labelling of MPC, MSB and/or MPB 1 mark for correct shift from MPC to MSC, divergent or parallel 1 mark for showing the external cost at market output Q ($MSC > MPC$) 1 mark for the new social optimum equilibrium (Q_1) identified where MSC equals MSB (Q_1) or overproduction 1 mark for welfare loss/gain identified 1 mark for quantity of waste water treatment Diagram required e.g. 	(4)

Question Number	Answer	Mark
(b)	Analysis 1 The only correct answer is C A not correct because the fine will increase efforts by the firm to reduce illegal discharges B not correct because the fine will increase costs and reduce profits D not correct because the intervention will not result in a net welfare loss.	(1)

Q11.

Question Number	Answer	Mark
(a)	Knowledge 2, Application 2, Analysis 1 Knowledge and analysis Up to 3 marks for: - the meaning of externalities, e.g. effect on third party, outside the transaction, difference between social cost and private cost (1). Points might include: - use of plastic which is hard to recycle/other single use plastic arguments (1) - HFSS food can cause obesity and have third party effects (1) - more congestion/air pollution from food deliveries (1) (positive) externality due to rise in productivity generated by cash rich, time poor professionals might allow people do impact third parties positively (1) Application: 2 marks for data (1+1 or 2): 1.4 billion containers (1) or other plastic items to allow food to travel safely e.g. heat retention (1) - over-consumption / order more food (1) e.g. impact on other NHS users (1) - HFSS food (1) - falling profit margins (1) or increased competition in food delivery might mean wages fall so fewer wider macro benefits (1) - don't even have to get up/lack of exercise of consumers ordering from their sofa (1) NB 2 marks may be awarded for a diagram e.g. a negative externalities in production of food that has to be delivered	(5)

Question Number	Answer	Mark
(b)	Knowledge 2, Application 2, Analysis 2, Evaluation 2 Knowledge/implicit understanding: (1+1) marks e.g. • demand is price inelastic means that the percentage change in price causes a smaller percentage change in quantity demanded • demand is income elastic means that percentage change in income causes a greater percentage change in demand Application: (1+1 or 2) e.g. • food delivery firms' charge of £5 is a very small percentage of consumer income so they might be seen as essential or basic goods (Ext B line 5) • three dominant firms Just Eat, Uber Eats and Deliveroo (Ext B line 2) • The food delivery industry has boomed in recent years (Ext B line 2) • cash rich, time poor (Ext B line 7) • income growth prior to 2020 meant delivery has boomed • HFSS addiction/habit forming • Convenient mobile app, so PED inelastic • 5 times more expensive than cooking at home Analysis: (1+1) development of each type of elasticity, e.g. • if demand is price inelastic then revenue will rise for the firm when it raises price • habit forming behaviour so PED inelastic • few close substitutes so PED inelastic • small part of overall consumer spending so PED inelastic • food delivery services are a very small percentage of consumer income so they might be seen as essential or basic goods • if food delivery is a luxury the demand likely to be income elastic, e.g. then the firms will benefit in times of economic growth, but lose demand in a recession • if food delivery is a treat in an economic downturn then it is an inferior good NB up to 2 marks may be awarded for diagram(s) (1+1)	
	Evaluation: (2 marks for any relevant point, or two points 1 +1) • the 30% of prices paid by restaurants is not seen by the consumer • depends on geographical areas e.g. rural vs urban areas. • the worldwide health crisis might have caused distortions in the data • third-party food delivery firms are inferior goods if demand rises when incomes fall • other factors affect demand apart from income, e.g., availability of substitutes (eating in restaurants is impossible in global health crisis) • discussion of normal vs inferior question e.g. compared to eating in a restaurant or cooking at home • income elastic vs. inelastic e.g. demand for third-party food delivery is price elastic for some income groups when incomes fall as it takes up a larger percentage of income	(8)

Question Number	Indicative content	Mark
(c)	Knowledge 2, Application 2, Analysis 4 Points might include: • integration includes horizontal/vertical/conglomerate joining of firms • leads to economies of scale • increased market share • removal of competition • increased monopoly power • international rivals can avoid taxes • prices higher for consumers • reduced choice for restaurants and consumers • monopsony power over restaurants • technical advantages of vertical integration • vertical power over firms can leave other firms with high barriers to entry • Just Eat has bought out rivals e.g. merger between Just Eat and Takeaway.com • Amazon has partially bought Deliveroo (16% so far) which may have saved it, and/or improved it on the tech side NB effects might include more than one types of economies of scale, or other combined analysis. Award diagrammatic analysis e.g. economies of scale LRAC	(8)

Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–2	Displays isolated or imprecise knowledge and understanding of terms, concepts, theories and models. Use of generic or irrelevant information or examples. Descriptive approach which has no chains of reasoning or links between causes and consequences.
Level 2	3–5	Displays elements of knowledge and understanding of economic principles, concepts and theories. Applies economic ideas and relates them to economic problems in context, although does not focus on the broad elements of the question. A narrow response; chains of reasoning are developed but the answer may lack balance.

Level 3	6–8	Demonstrates accurate knowledge and understanding of the concepts, principles and models. Ability to link knowledge and understanding in context using relevant and focused examples which are fully integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems. The answer demonstrates logical and coherent chains of reasoning.
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Question Number	Indicative content	Mark
(c) continued	Evaluation 4 • mergers might be necessary to ensure the survival of firms (e.g. Deliveroo case) • further mergers might be limited by the CMA • supernormal profits might lead to lower prices, better technology and/or a more consumer-led market • prices may fall due to fall in average costs due to e.g. economies of scale • diseconomies of scale	(4)

Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–2	Identification of generic evaluative comments without supporting evidence/reference to context. No evidence of a logical chain of reasoning.
Level 2	3–4	Evaluative comments supported by relevant reasoning and appropriate reference to the context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Question Number	Indicative content	Mark
(d)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 Microeconomic effects of depreciation may include: • higher cost of imports of food and therefore higher cost of production for hospitality sector • lower profits for firms in hospitality • higher prices of food for consumers • consumers have lower standards of living if they are on fixed incomes/ • falling real incomes so demand for restaurant meals and food delivery services might shift left • workers have higher incomes if working in exporting firms in other industries, so might use more delivery services • restaurants might begin to use more locally produced food Macroeconomic effects of depreciation may include: • capital in-flows, e.g. benefits to multinational firms such as Just Eat for investment in the US using other currencies/ AD and/or LRAS shift out • cost push inflation/SRAS shifts in • more injections and fewer withdrawals leading to rise in AD and multiplier effect • increased employment • incentive to use transfer pricing • UK tourism might be boosted by depreciation which would improve local restaurants NB for a Level 4 response there must be micro and macro effects and use of data with reference to restaurants or other food delivery services. Higher costs, changes in competitiveness, environmental changes could be seen as micro or macro, for example, can be used as either micro or macroeconomics. NB Effects in KAA could be used as evaluation and vice-versa.	(25)

	9 marks for evaluation – points might include: • depends on the nature of the restaurant. Some try to use locally sourced ingredients • the pound has recovered significantly since 2016 • food costs are a small cost relative to other costs, e.g. rent and delivery • some restaurants are specialised in UK food so cost effect very low • other factors are more important, e.g. the heavy fiscal spending in response to the global health crisis • other factors determining capital inflows • relative depreciation or trade weighting of other currencies • offsetting costs with altering portion sizes or changing menu options • impact on tourism small relative to the increase in the production of food • regional differences with the UK e.g. whether there is a local food supply option • Marshall Lerner • J-curve	
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Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays ability to apply knowledge in context but will focus on small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models.
Level 2	5–8	Shows ability to apply economic ideas and relate them to economic problems in context. Displays knowledge and understanding of economic principles, concepts and theories to make limited analysis or narrow analysis.
Level 3	9–12	Analysis is clear and coherent with evidence well integrated, although may focus on some of the broad elements of the question more than others. Shows ability to apply economic ideas and relate them directly to the broad elements in the question.
Level 4	13–16	Analysis is relevant, clear and coherent with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic effects. A clear understanding of economic principles, concepts, theories and arguments.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of evaluative comments without explanation.
Level 2	4–6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to the context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Question Number	Indicative content	Mark
(e)	Knowledge 4, Application 4, Analysis 8, Evaluation 9 16 marks for KAA, for effects of increased minimum wage Microeconomic effects may include: • rise in wages will benefit very low paid workers. Might be shown on increasing NMW diagram • effect on cost to firms e.g. reduced profit margins. Might be shown on a cost and revenue diagram. • excess supply of hospitality workers • effect on prices consumer surplus falls, output falls, price increases • impact on quantity of labour demanded/contraction of labour demand • impact on productivity • inefficient resource allocation due to distortion of price mechanism Macroeconomic effects may include: • income equality improvements – may use Gini/Lorenz • prices rise – inflationary effects • reduction in relative poverty • AD: increase in consumption as incomes increase/fall in investment as profits fall • SRAS: cost-push • LRAS: if investment falls, then LRAS shifts in • reduced employment • higher income tax revenue • lower unemployment related benefits NB for a Level 4 response there must be micro and macro effect (s) and use of data with reference to restaurants or other food delivery services. Employment or welfare issues could be seen as micro or macro, for example, can be used as either micro or macroeconomics. NB Effects in KAA could be used as evaluation and vice-versa.	(25)

	9 marks for evaluation – points might include: • effect on firms depends on how much wage costs are as a proportion of overall costs • impact on real output depends on level of spare capacity/elasticity of LRAS/magnitude of multiplier • effect on workforce of employment regulation depends on elasticity for demand for labour, and degree with which it can be replaced with capital • depends on PED of food delivery firm market as a whole • consumers might also have increases in wages so profits might not fall • some firms will face decreased demand and go out of business. Loss of jobs and other economic activity • depends on the PED and PES in factor market and product market as a whole • LRAS might rise as economically inactive workers enter labour force to look for jobs • higher tax revenue may not occur if unemployment increases or investment decreases • higher unemployment related benefits if unemployment increase • productivity/motivation arguments	
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Knowledge, application and analysis		
Level	Mark	Descriptor
	0	A completely inaccurate response.
Level 1	1–4	Displays ability to apply knowledge in context but will focus on small range of elements. Demonstrates understanding by identifying relevant information. Demonstrates knowledge and understanding of terms, concepts, theories and models.
Level 2	5–8	Shows ability to apply economic ideas and relate them to economic problems in context. Displays knowledge and understanding of economic principles, concepts and theories to make limited analysis or narrow analysis.
Level 3	9–12	Analysis is clear and coherent with evidence well integrated, although may focus on some of the broad elements of the question more than others. Shows ability to apply economic ideas and relate them directly to the broad elements in the question.
Level 4	13–16	Analysis is relevant, clear and coherent with evidence fully and reliably integrated. Economic ideas are carefully selected and applied appropriately to economic issues and problems covering both microeconomic and macroeconomic effects. A clear understanding of economic principles, concepts, theories and arguments.

Evaluation		
Level	Mark	Descriptor
	0	No evaluative comments.
Level 1	1–3	Identification of evaluative comments without explanation.
Level 2	4–6	Evaluative comments with limited explanations. Evidence of evaluation of alternative approaches which is generic or unbalanced leading to limited judgements.
Level 3	7–9	Evaluative comments supported by relevant reasoning and appropriate reference to the context. Evaluation recognises different viewpoints and is critical of the evidence provided and/or the assumptions underlying the analysis enabling informed judgements to be made.

Q12.

Question Number	Answer	Mark
(a)	Knowledge 1 Knowledge/understanding 1 mark for definition, e.g. This occurs when government intervention in the economy causes a net welfare loss/decline in economic welfare.	(1)

Question Number	Answer	Mark
(b)	Knowledge 2 Knowledge 1 mark for reason and 1 for linked development Possible answers include: • Positive externalities (1) resulting from people using railways rather than roads (1) • Risk/imperfect information (1) if the project is undertaken by the private sector it may be less likely to happen (1)	(2)

Question Number	Answer	Mark
(c)	The only correct answer is A <i>B is not correct because this would cause the demand for rail travel to shift to the left</i> <i>C is not correct because this would cause the supply curve to shift to the right</i> <i>D is not correct because this would cause the supply curve to shift to the right</i>	(1)